

Active Learning for Joint Communication and Sensing

1 Measurement Model

- The base station (BS) has N_{Tx} transmit antennas and N_{Rx} receive antennas for radar sensing that are co-located and arranged in a uniform linear array. There are K single-antenna communication users and a single radar point target.
- At time $t = 1, 2, \dots, T$, the base station transmits $\mathbf{x}_t \in \mathbb{C}^{N_{\text{Tx}}}$ where $[\mathbf{x}_t]_i$ is the discrete sample of the waveform transmitted by the i^{th} antenna, $i = 1, \dots, N_{\text{Tx}}$. Following [1, 2], we can consider

$$\mathbf{x}_t = \underbrace{\mathbf{V}_t^{(c)} \mathbf{c}_t}_{\text{comm. part}} + \mathbf{V}_t^{(s)} \mathbf{s}_t, \quad t = 1, \dots, T \quad (1)$$

where we have

- Communication data $\mathbf{c}_t \in \mathbb{C}^K$ where $[\mathbf{c}_t]_k$ = communication symbol for k^{th} user, $k = 1, \dots, K$. The communication data $\mathbf{c}_t$ cannot be controlled by the BS.
- We can design the matrix $\mathbf{V}_t^{(c)} \in \mathbb{C}^{N_{\text{Tx}} \times K}$ and the sensing vector $\mathbf{V}_t^{(s)} \mathbf{s}_t \in \mathbb{C}^{N_{\text{Tx}}}$. Further, we can consider $\mathbf{s}_t \sim \mathcal{CN}(0, \mathbf{I})^1$ as done in [1, 2, 3] and focus on designing the matrix $\mathbf{V}_t^{(s)} \in \mathbb{C}^{N_{\text{Tx}} \times N_{\text{Tx}}}$.
- The signal received at time t by the K communication users is denoted by vector $\mathbf{y}_t^{(c)} \in \mathbb{C}^K$ and the BS receives radar signal $\mathbf{y}_t^{(s)} \in \mathbb{C}^{N_{\text{Rx}}}$ given by

$$\mathbf{y}_t^{(c)} = \mathbf{H} \mathbf{x}_t + \mathbf{n}_t^{(c)} \quad (2)$$

$$\mathbf{y}_t^{(s)} = \mathbf{G} \mathbf{x}_t + \mathbf{n}_t^{(s)} \quad (3)$$

where $\mathbf{H} = [\mathbf{h}_1 \ \dots \ \mathbf{h}_K]^H \in \mathbb{C}^{K \times N_{\text{Tx}}}$ is the communication channel matrix, $\mathbf{G} \in \mathbb{C}^{N_{\text{Rx}} \times N_{\text{Tx}}}$ is the radar target response matrix, $\mathbf{n}_t^{(c)} \sim \mathcal{CN}(0, \sigma_c^2 \mathbf{I})$, $\mathbf{n}_t^{(s)} \sim \mathcal{CN}(0, \sigma_s^2 \mathbf{I})$ are i.i.d noise at the antennas.

- The BS receives feedback signal $\mathbf{y}_t^{(s)}$ from the radar target
- BS cannot observe $\mathbf{y}_t^{(c)}$ and the k^{th} user can only observe $[\mathbf{y}_t^{(c)}]_k$ for $k = 1, \dots, K$.
- The target response matrix $\mathbf{G}$ in eq. (3) is given by

$$\mathbf{G} = \alpha \mathbf{a}_{\text{Rx}}(\theta) \mathbf{a}_{\text{Tx}}^H(\theta) \quad (4)$$

where $\alpha \in \mathbb{C}$ is the reflection coefficient and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ is the angle corresponding to the target location. The array response vector $\mathbf{a}_{\text{Rx}}(\theta) \in \mathbb{C}^{N_{\text{Rx}}}$ is given by (for even N_{Rx})² [1]

$$\mathbf{a}_{\text{Rx}}(\theta) = [1 \quad e^{j\pi \sin(\theta)} \quad e^{j\pi 2 \sin(\theta)} \quad \dots \quad e^{j\pi (N_{\text{Rx}}-1) \sin(\theta)}]^T \quad (5)$$

Similarly, $\mathbf{a}_{\text{Tx}}(\theta) \in \mathbb{C}^{N_{\text{Tx}}}$ can be defined for even N_{Tx} .

- The BS is assumed to know the channel matrix $\mathbf{H}$. Note that $\mathbf{H}$ and $\mathbf{G}$ can share common parameters, although this is not always exploited in the literature.

¹ $\mathcal{CN}(\mu, \Gamma)$ is the circularly symmetric complex Gaussian distribution

²Taking the origin at the center of the array and assuming the adjacent array elements are placed at $d = \frac{\lambda}{2}$ where λ is the wavelength of the carrier

1.1 Communication Constraints

- The designed waveform needs to obey the following communication constraints:

– **Power constraint:** The average power constraint at time $t = 1, \dots, T$ can be written as:

$$\mathbb{E} \left[\|\mathbf{x}_t\|_2^2 \right] = \text{tr}((\mathbf{V}_t^{(c)})^H \mathbf{V}_t^{(c)} + (\mathbf{V}_t^{(s)})^H \mathbf{V}_t^{(s)}) \leq P_T \quad (6)$$

– **SINR constraint:** Let $\mathbf{V}_t^{(c)} = [\mathbf{v}_{1,t} \ \mathbf{v}_{2,t} \ \dots \ \mathbf{v}_{K,t}]$ and let SINR for k^{th} user at time t be $\gamma_{k,t}$ defined as follows

$$\gamma_{k,t} = \frac{|\mathbf{h}_i^H \mathbf{v}_{i,t}|^2}{\sum_{j=1, j \neq i}^K |\mathbf{h}_i^H \mathbf{v}_{j,t}|^2 + \|\mathbf{h}_i^H \mathbf{V}_t^{(s)}\|^2 + \sigma_C^2}, \quad k = 1, \dots, K \quad (7)$$

then the SINR constraint is written as

$$\gamma_{k,t} \geq \Gamma_k, \quad k = 1, \dots, K \quad (8)$$

– Further communication constraints can be added, like constraint to enable coherent decoding at the user side.

- Note that the problem of sensing the radar target is sequential in nature even without the communication constraints.

2 Online learning for the single-user single-target cases

2.1 Assumptions to simplify the problem

1. We only have single user and single target;
2. $N_{\text{Rx}} = N_{\text{Tx}}$ ³

2.2 Problem Setting

2.2.1 Joint sensing and communication model

At t time, we send out

The sensing target response

$$\mathbf{y}_t^{(s)} = \alpha \mathbf{a}_{\text{Rx}}(\theta) \mathbf{a}_{\text{Tx}}^H(\theta) \mathbf{x}_t + \mathbf{n}_t^{(s)}$$

where $\theta \in [-\pi/2, \pi/2]$ is the target angle,

$$\mathbf{a}_{\text{Rx}}(\theta) = [1 \quad e^{j\pi \sin(\theta)} \quad e^{j2\pi \sin(\theta)} \quad \dots \quad e^{j\pi(N_{\text{Rx}}-1) \sin(\theta)}]^T \quad (9)$$

Similarly, $\mathbf{a}_{\text{Tx}}(\theta) \in \mathbb{C}^{N_{\text{Tx}}}$ can be defined for even N_{Tx} .

- Related responses of joint sensing and communication,
 1. Here, we discretized θ in to a N -grid in $[-\pi/2, \pi/2]$, denoted as $\theta_1, \theta_2, \dots, \theta_n$.
 2. The communication constraints

$$|\mathbf{h}^H \mathbf{w}_t|^2 \geq \Gamma \sigma_c^2, \quad \|\mathbf{w}_t\|^2 \leq P_T$$

where Γ is the required SINR level, and P_T is the transmit power budget.

³The case $N_{\text{Rx}} > N_{\text{Tx}}$ can be addressed by defining $\tilde{\mathbf{x}}_t := [\mathbf{x}_t \ \mathbf{0}]$ with $N_{\text{Rx}} - N_{\text{Tx}}$ no. of zeros and calculating $|\tilde{\mathbf{x}}_t^H \mathbf{y}_t^{(s)}|$ instead

- **Rewards and regret.** every time we select $\mathbf{w}_t$. and generate $\mathbf{x}_t$ according to

$$\mathbf{x}_t = \mathbf{w}_t c_t \quad (10)$$

where $c_t \in \mathbb{C}$ is the communication symbol transmitted at time t and $\mathbf{w}_t \in \mathbb{C}^{N_{\text{Tx}}}$ is the beamforming vector that needs to be designed.

The objective function is to optimize CRB, which is

$$r(\mathbf{w}) = \text{CRB}(\mathbf{w}; \theta) = \frac{\sigma_s^2}{2|\alpha|^2 T \|\dot{\mathbf{a}}_{\text{Rx}}(\theta)\|^2 |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}|^2} \quad (11)$$

where $\dot{\mathbf{a}}_{\text{Rx}}(\theta) = \frac{\partial}{\partial \theta} \mathbf{a}_{\text{Rx}}(\theta)$. The only observation we have is the $\mathbf{y}_t^{(s)}$,

$$\mathbf{y}_t^{(s)} = \alpha \mathbf{a}_{\text{Rx}}(\theta) \mathbf{a}_{\text{Tx}}^H(\theta) \mathbf{x}_t + \mathbf{n}_t^{(s)}$$

We also observe that, for a fixed θ , minimising $\text{CRB}(\mathbf{w}; \theta)$ w.r.t $\mathbf{w}$ is equivalent to maximising $f(\mathbf{w}; \theta) := |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}|^2$. However, we cannot directly sample $f(\mathbf{w}; \theta)$ since the influence of θ can only be observed via the measurement $\mathbf{y}_t^{(s)}$.

Let $N_{\text{Rx}} = N_{\text{Tx}}^4$, then we have

$$\begin{aligned} r_1(\mathbf{w}_t) &= \left| \mathbf{x}_t^H \mathbf{y}_t^{(s)} \right| \\ &= \left| \mathbf{x}_t^H (\mathbf{G} \mathbf{x}_t + \mathbf{n}_t^{(s)}) \right| \\ &= \left| \mathbf{x}_t^H \mathbf{G} \mathbf{x}_t + \mathbf{x}_t^H \mathbf{n}_t^{(s)} \right| \\ &= \left| \alpha |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{x}_t|^2 + \mathbf{x}_t^H \mathbf{n}_t^{(s)} \right| \\ &= \left| \alpha |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}_t|^2 + \mathbf{x}_t^H \mathbf{n}_t^{(s)} \right| \quad (\text{as } \mathbf{x}_t = \mathbf{w}_t c_t \text{ and } |c_t| = 1) \end{aligned}$$

- Here, output of $g_1(\mathbf{x}_t, \mathbf{y}_t^{(s)})$ is always positive. However, it does not provide an unbiased estimate of $|\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}_t|^2$ as $\mathbb{E} \left[g_1(\mathbf{x}_t, \mathbf{y}_t^{(s)}) \right] \neq |\alpha| |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}_t|^2$

Another possible surrogate function to $f(\mathbf{w}_t; \theta)$ could be

$$r_2(\mathbf{w}_t) = \Re(\mathbf{x}_t^H \mathbf{y}_t^{(s)}) = \Re(\alpha) |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}_t|^2 + \Re(\mathbf{x}_t^H \mathbf{n}_t^{(s)})$$

where $\Re(z)$ = real part of z .

- In this case, $\mathbb{E} \left[g_2(\mathbf{x}_t, \mathbf{y}_t^{(s)}) \right] = \Re(\alpha) |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}_t|^2$, however output of $g_2(\mathbf{x}_t, \mathbf{y}_t^{(s)})$ need not be positive.
- We can further discuss on what surrogate function would make sense.

2.3 Transmit beamformer design

Following [1], the transmit beamformer $\mathbf{w}$ is obtained by solving the following optimisation problem:

$$\begin{aligned} \max_{\mathbf{w}} \quad & |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}|^2 \\ \text{s.t.} \quad & |\mathbf{h}^H \mathbf{w}|^2 \geq \Gamma \sigma_c^2, \quad \|\mathbf{w}\|^2 \leq P_T \end{aligned} \quad (12)$$

⁴The case $N_{\text{Rx}} > N_{\text{Tx}}$ can be addressed by defining $\tilde{\mathbf{x}}_t := [\mathbf{x}_t \mathbf{0}]$ with $N_{\text{Rx}} - N_{\text{Tx}}$ no. of zeros and calculating $|\tilde{\mathbf{x}}_t^H \mathbf{y}_t^{(s)}|$ instead

Consider the solution to the above optimization is $\mathbf{w}^*$. Then $\mathbf{w}^*$ can be written as $\mathbf{w}^* = a_1 \hat{\mathbf{u}} + a_2 \hat{\mathbf{v}}$ where the unit vectors $\hat{\mathbf{u}} \in \text{span}\{\mathbf{a}_{\text{Tx}}(\theta), \mathbf{h}\}$ and $\hat{\mathbf{v}} \in \text{span}\{\mathbf{a}_{\text{Tx}}(\theta), \mathbf{h}\}^\perp$. Further, $\|\mathbf{w}^*\|_2^2 \leq P_T \implies |a_1|^2 + |a_2|^2 \leq P_T$. We can show that $|a_2|^2 = 0$ and $|a_1|^2 = P_T$ by noticing that if not, then $\mathbf{w}^*$ is not optimal since $\mathbf{w}' = \sqrt{P_T} \frac{a_1}{|a_1|} \hat{\mathbf{u}}$ will have $|\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}'|^2 = P_T |\mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{u}}|^2 > |a_1|^2 |\mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{u}}|^2 = |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}^*|^2$ (and $\mathbf{w}'$ also satisfies the SINR constraint).

If $\sqrt{\frac{P_T}{N_{\text{Tx}}}} \mathbf{a}_{\text{Tx}}(\theta)$ satisfies the communication constraint⁵, i.e., $\frac{P_T}{N_{\text{Tx}}} |\mathbf{h}^H \mathbf{a}_{\text{Tx}}(\theta)|^2 \geq \Gamma \sigma_c^2$ then we just take $\mathbf{w}^* = \sqrt{\frac{P_T}{N_{\text{Tx}}}} \mathbf{a}_{\text{Tx}}(\theta)$. In this case, the communication constraint is “inactive”.

Otherwise, the communication constraint is “active”, in which case the optimal beamformer $\mathbf{w}^* \in \text{span}\{\mathbf{a}_{\text{Tx}}(\theta), \mathbf{h}\}$ is of the form [1]

$$\mathbf{w}^* = \tilde{\beta}_1 \mathbf{h} + \tilde{\beta}_2 \mathbf{a}_{\text{Tx}}(\theta) = \beta_1 \hat{\mathbf{h}} + \beta_2 \hat{\mathbf{a}}_{\mathbf{h}\perp}$$

where $\beta_1, \beta_2 \in \mathbb{C}$, $\hat{\mathbf{h}} = \frac{\mathbf{h}}{\|\mathbf{h}\|_2}$ and $\hat{\mathbf{a}}_{\mathbf{h}\perp} = \frac{(\mathbf{I} - \hat{\mathbf{h}}\hat{\mathbf{h}}^H) \mathbf{a}_{\text{Tx}}(\theta)}{\|(\mathbf{I} - \hat{\mathbf{h}}\hat{\mathbf{h}}^H) \mathbf{a}_{\text{Tx}}(\theta)\|_2}$ (i.e., the projection of $\mathbf{a}_{\text{Tx}}(\theta)$ on nullspace of $\hat{\mathbf{h}}$ normalised to unit norm).

From the total power constraint, we have $\|\mathbf{w}^*\|_2^2 = |\beta_1|^2 + |\beta_2|^2 = P_T$. To satisfy communication constraint, we can set

$$|\mathbf{h}^H \mathbf{w}^*|^2 = |\beta_1|^2 \|\mathbf{h}\|_2^2 = \Gamma \sigma_c^2 \implies |\beta_1| = \sqrt{\frac{\Gamma \sigma_c^2}{\|\mathbf{h}\|_2^2}}, |\beta_2| = \sqrt{P_T - |\beta_1|^2}$$

Let $\beta_1 = |\beta_1| e^{j\phi_1}$, $\beta_2 = |\beta_2| e^{j\phi_2}$, then ϕ_1, ϕ_2 should be set to maximise $|\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}^*|^2 = |\beta_1 \mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{h}} + \beta_2 \mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{a}}_{\mathbf{h}\perp}|^2$. This can be done by setting $\phi_1 = 0$ and $\phi_2 = \angle(\mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{h}}) - \angle(\mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{a}}_{\mathbf{h}\perp})$.⁶

2.4 Transmit beamforming in multiple directions

Let $\mathcal{G} = \{\theta_1, \dots, \theta_{N_g}\} \subset [-\frac{\pi}{2}, \frac{\pi}{2})$ be a discretised angular grid such that for all $\theta_i, \theta_j \in \mathcal{G}, i \neq j$ we have $\mathbf{a}_{\text{Tx}}(\theta_i)^H \mathbf{a}_{\text{Tx}}(\theta_j) = 0$. Let $\mathbf{w}_i^*$ be the solution of (14) for angle $\theta_i \in \mathcal{G}$. Suppose we want to probe K gridpoints corresponding to indices $\mathcal{I} \subset [N_g], |\mathcal{I}| = K$ simultaneously, then one approach can be to use

$$\mathbf{w} = \frac{1}{K} \sum_{k \in \mathcal{I}} \mathbf{w}_k^*$$

which distributes the sensing power equally in the K directions. Note that $\mathbf{w}$ satisfies the communication constraint as the component of $\mathbf{w}_k^*, k = 1, \dots, K$ in the direction of $\mathbf{h}$ are equal and in-phase.

In general, to probe K angles $\theta_1, \dots, \theta_K \in [-\frac{\pi}{2}, \frac{\pi}{2})$, we can formulate the problem as follows. Let $\Theta = \{\theta_1, \dots, \theta_K\}$ and $\mathbf{A}_\Theta = [\mathbf{a}_{\text{Tx}}(\theta_1), \dots, \mathbf{a}_{\text{Tx}}(\theta_K)]$, then

$$\begin{aligned} \max_{\mathbf{w}} \quad & \sum_{\theta \in \Theta} |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}|^2 = \|\mathbf{A}_\Theta^H \mathbf{w}\|_2^2 = \mathbf{w}^H \mathbf{A}_\Theta \mathbf{A}_\Theta^H \mathbf{w} \\ \text{s.t.} \quad & |\mathbf{h}^H \mathbf{w}|^2 \geq \Gamma \sigma_c^2, \|\mathbf{w}\|^2 \leq P_T \end{aligned} \quad (13)$$

Without the communication constraint, the solution is given by obtaining eigen-decomposition of $\mathbf{A}_\Theta \mathbf{A}_\Theta^H = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H$ and setting $\mathbf{w}^* = \sqrt{P_T} \mathbf{u}_1$ where $\mathbf{u}_1$ is the unit-norm eigenvector of $\mathbf{A}_\Theta \mathbf{A}_\Theta^H$ corresponding to the maximum eigenvalue. To accomodate the communication constraint, we can think of a similar approach as above by taking $\mathbf{w}^* = \tilde{\beta}_1 \mathbf{h} + \tilde{\beta}_2 \mathbf{u}_1$ (need to check if this will indeed be optimal). This is similar to the Theorem 3 in [1], however, their optimisation problem is slightly different.

To probe a angular region $\Theta \subset [-\frac{\pi}{2}, \frac{\pi}{2})$, we can formulate the beamformer design as the following optimisation problem:

$$\begin{aligned} \max_{\mathbf{w}} \min_{\theta \in \Theta} \quad & |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}|^2 \\ \text{s.t.} \quad & |\mathbf{h}^H \mathbf{w}|^2 \geq \Gamma \sigma_c^2, \|\mathbf{w}\|^2 \leq P_T \end{aligned} \quad (14)$$

⁵This implies the user and the target are in the same direction

⁶In [1], they use $\phi_1 = -\angle(\mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{h}})$ and $\phi_2 = -\angle(\mathbf{a}_{\text{Tx}}(\theta)^H \hat{\mathbf{a}}_{\mathbf{h}\perp})$

2.5 Transmit beamforming strategies (without communication constraints)

Let $\mathcal{G} = \{\theta_1, \dots, \theta_{N_g}\} \subset [-\frac{\pi}{2}, \frac{\pi}{2})$ be a the angular grid such that for all $\theta_i, \theta_j \in \mathcal{G}, i \neq j$ we have $\mathbf{a}_{\text{Tx}}(\theta_i)^H \mathbf{a}_{\text{Tx}}(\theta_j) = 0$. This can be obtained as follows:

$$\begin{aligned} \mathbf{a}_{\text{Tx}}(\theta_i)^H \mathbf{a}_{\text{Tx}}(\theta_j) &= \sum_{n=0}^{N_{\text{Tx}}-1} e^{j\pi n(\sin(\theta_j) - \sin(\theta_i))} \\ &= \frac{e^{j\pi N_{\text{Tx}}(\sin(\theta_j) - \sin(\theta_i))} - 1}{e^{j\pi(\sin(\theta_j) - \sin(\theta_i))} - 1} \\ &= e^{j\pi(\frac{N_{\text{Tx}}-1}{2})(\sin(\theta_j) - \sin(\theta_i))} \frac{\sin(\frac{\pi}{2}N(\sin(\theta_j) - \sin(\theta_i)))}{\sin(\frac{\pi}{2}(\sin(\theta_j) - \sin(\theta_i)))} \end{aligned}$$

Thus, we can choose the grid such that $\sin(\theta_i) = -1 + \frac{2(i-1)}{N_{\text{Tx}}}, i = 1, \dots, N_{\text{Tx}}$.

Let $\mathbf{A} = \frac{1}{\sqrt{N_{\text{Tx}}}} [\mathbf{a}_{\text{Tx}}(\theta_1) \quad \mathbf{a}_{\text{Tx}}(\theta_2) \quad \dots \quad \mathbf{a}_{\text{Tx}}(\theta_{N_{\text{Tx}}})]$ then $\mathbf{A}$ will be a unitary matrix. Suppose that we want beamformer $\mathbf{w}$ to have maximum average gain over a subset $\Theta \subset [N_{\text{Tx}}]$.

$$\max_{\mathbf{w} \in \mathbb{C}^{N_{\text{Tx}}}} \sum_{i \in \Theta} |\mathbf{a}_{\text{Tx}}(\theta_i)^H \mathbf{w}|^2 = \|\mathbf{A}_{\Theta}^H \mathbf{w}\|_2^2 \text{ such that } \|\mathbf{w}\|_2^2 = 1$$

We can write $\mathbf{w} = \mathbf{A}_{\Theta} \mathbf{x}_1 + \mathbf{A}_{\Theta^c} \mathbf{x}_2$ which implies $\|\mathbf{A}_{\Theta}^H \mathbf{w}\|_2^2 = \|\mathbf{x}_1\|_2^2$ and $\|\mathbf{w}\|_2^2 = \|\mathbf{A}_{\Theta} \mathbf{x}_1 + \mathbf{A}_{\Theta^c} \mathbf{x}_2\|_2^2 = \|\mathbf{x}_1\|_2^2 + \|\mathbf{x}_2\|_2^2$. Thus, we have the following equivalent optimisation problem

$$\max_{\mathbf{x}_1, \mathbf{x}_2} \|\mathbf{x}_1\|_2^2 \text{ such that } \|\mathbf{x}_1\|_2^2 + \|\mathbf{x}_2\|_2^2 = 1$$

Thus, any vector $\mathbf{x}_1$ such that $\|\mathbf{x}_1\|_2^2 = 1$ will give the optimal $\mathbf{w} = \mathbf{A}_{\Theta} \mathbf{x}_1$. This implies, both $\mathbf{w} = \mathbf{a}_{\text{Tx}}(\theta_i), i \in \Theta$ and $\mathbf{w} = \frac{1}{|\Theta|} \sum_{i \in \Theta} \mathbf{a}_{\text{Tx}}(\theta_i)$ maximise the average gain.

Algorithm 2 Online Joint Communication and Sensing

Require: Initialize dataset $D_0 = \{\}$, $\mathcal{C}_0 \subset [-\frac{\pi}{2}, \frac{\pi}{2})$

- 1: **for** time step $t = 1, 2, \dots$ **do**
 - 2: Select $\mathbf{w}_t$ which covers $\mathcal{C}_{t-1}$
 - 3: Send out $\mathbf{x}_t = \mathbf{w}_t c_t$ and observe $\mathbf{y}_t = \mathbf{G}(\theta) \mathbf{x}_t + \mathbf{n}_t^{(s)}$. Collect data $D_t = D_{t-1} \cup \{\mathbf{x}_t, \mathbf{y}_t\}$.
 - 4: Estimate $\hat{\theta}_t$ based on D_t (using MUSIC or ESPRIT algorithms)
 - 5: Construct confidence set $\mathcal{C}_t$. For example, MUSIC spectrum can be thresholded for obtaining a confidence set
 - 6: **end for**
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[Haitong: TODO: Implementing the algorithm and think about the UCB design of the problem.]

[Haitong: Piya and Part mentioned that instead of only beamform on only one direction θ_i , we can also beamform on two directions. We can think about how this can help explorations.]

[Haitong: An immediate question is that what will the observations $\mathbf{y}_t^{(s)}$ tell us if you]

3 Simulation Results

References

- [1] F. Liu, Y.-F. Liu, A. Li, C. Masouros, and Y. C. Eldar, "Cramér-Rao Bound Optimization for Joint Radar-Communication Beamforming," *IEEE Transactions on Signal Processing*, vol. 70, pp. 240–253, 00 2022.

Algorithm 1 Online Joint Communication and Sensing [Haitong: discretized θ , UCB version]

Require: Initialize dataset $D_0 = \{\}$, discretized θ grid $\theta_1, \theta_2, \theta_3, \dots, \theta_N$, initialize visit counters $T_i(0) = 0$ for $i = 1, 2, \dots, N$.

- 1: For each discrete angle θ_i , solve the optimal $\mathbf{w}_i$ to minimize the CRB with respect to SINR constraints,

$$\begin{aligned} \max_{\mathbf{w}_i} & |\mathbf{a}_{\text{Tx}}(\theta)^H \mathbf{w}|^2 \\ \text{s.t.} & |\mathbf{h}^H \mathbf{w}_i|^2 \geq \Gamma \sigma_c^2 \\ & \|\mathbf{w}_i\|^2 \leq P_T \end{aligned} \quad (15)$$

[Haitong: For first-step implementation we can go ahead to use the solution in [1].]

- 2: **for** time step $t = 1, 2, \dots$ **do**
- 3: Update upper confidential bound

$$\text{UCB}_t(\theta_i) = \begin{cases} \infty & \text{if } T_i(t-1) = 0 \\ \hat{r}_i(t-1) + \sqrt{\frac{2 \log(1/\delta)}{T_i(t-1)}} & \text{otherwise.} \end{cases} \quad (16)$$

[Haitong: Check if the reward distribution is sub-Gaussian, if it is ,we can go ahead to apply the regret results, probably with additional linear term due to the discretization.]

- 4: Select $\mathbf{w}_t = \mathbf{w}_i$, where $\mathbf{w}_i$ corresponds to the θ_i maximize the $\text{UCB}_t(\theta_i)$.
- 5: Send out $\mathbf{x}_t = \mathbf{w}_t c_t$ and observe $\mathbf{y}_t = \mathbf{G}(\theta) \mathbf{x}_t + \mathbf{n}_t^{(s)}$. Collect data $D_t = D_{t-1} \cup \{\mathbf{x}_t, \mathbf{y}_t\}$.
- 6: Calculate the reward by the surrogate function

$$\tilde{r}(\mathbf{w}_t) = \left| \mathbf{x}_t^H \mathbf{y}_t^{(s)} \right| ,$$

Update the mean reward $\hat{r}_i(t)$ and visit count $T_i(t)$ for each θ_i .

- 7: **end for**
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- [2] X. Liu, T. Huang, N. Shlezinger, Y. Liu, J. Zhou, and Y. C. Eldar, “Joint Transmit Beamforming for Multiuser MIMO Communications and MIMO Radar,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 3929–3944, 2019.
- [3] K. M. Attiah and W. Yu, “Active Beamforming for Integrated Sensing and Communication,” *2023 IEEE International Conference on Communications Workshops (ICC Workshops)*, vol. 00, pp. 1469–1474, 2023.